

Mehul Damani

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Education

- Ph.D. Massachusetts Institute of Technology (MIT)** Aug 2022 – present
Ph.D. candidate in Computer Science
- **Advisor:** [Jacob Andreas](#) ; **GPA:** 4.9/5.0
- B.Eng Nanyang Technological University (NTU)** Aug 2018 – May 2022
Bachelor's in Mechanical Engineering
- **GPA:** 4.74/5.0 ; Honours (Highest Distinction)

Experience

- MIT-IBM Watson AI Lab** Cambridge, MA
Research Intern May 2025 – Aug 2025
- Reinforcement Learning for LLM tool use.
- New York University** Remote
Research Intern (Advised by [Lerrel Pinto](#)) Jan 2021 – Aug 2022
- Developed curriculum generation method that improved performance by upto 20% on robotic tasks.
- National University of Singapore** Singapore
Research Intern (Advised by [Guillaume Sartoretti](#)) Apr 2020 – July 2022
- Developed decentralized reinforcement learning (RL) methods for multi-agent systems.
 - Co-authored 4 papers, open-sourced code with 100+ stars on Github.
 - Achieved fourth position in the RL track of the NeurIPS 2020 Flatland Challenge.
- Satellite Research Centre (NTU)** Singapore
Research Intern Sept 2019 – Apr 2020
- Developed regression models to characterize drift and bias of sensors for integration into ADCS of a satellite.

Publications

- [12] **Self-Distillation Enables Continual Learning** 2026
 Idan Shenfeld, **Mehul Damani**, Jonas Hubotter, Pulkit Agrawal
International Conference on Machine Learning (ICML)
ICLR 2026 Workshop on Lifelong Agents (LLA) (Oral)
- [11] **Escaping the Mode: Multi Answer Reinforcement Learning in LMs** 2026
 Isha Puri, **Mehul Damani**, Idan Shenfeld, Marzyeh Ghassemi, Jacob Andreas, Yoon Kim
International Conference on Machine Learning (ICML)
- [10] **Beyond Binary Rewards: Training LMs to Reason about their Uncertainty** 2026
Mehul Damani, Isha Puri, Stewart Slocum, Idan Shenfeld, Leshem Choshen, Yoon Kim, Jacob Andreas
International Conference on Learning Representations (ICLR)
COLM 2025 Workshop on Test-Time Scaling and Reasoning Models (Oral)
- [9] **The Surprising Effectiveness of Test-Time Training for Abstract Reasoning** 2025
 Ekin Akyurek, **Mehul Damani**, Linlu Qiu, Han Guo, Yoon Kim, Jacob Andreas
International Conference on Machine Learning (ICML)
- [8] **Learning How Hard to Think: Input-Adaptive Allocation of LM Computation** 2025
Mehul Damani, Idan Shenfeld, Andi Peng, Andreea Bobu, Jacob Andreas

[7] **Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback** 2023

Stephen Casper, Xander Davies, Claudia Shi, Thomas Krendl Gilbert, Jérémy Scheurer, Javier Rando, Rachel Freedman, Tomasz Korbak, David Lindner, Pedro Freire, Tony Wang, Samuel Marks, Charbel-Raphaël Segerie, Micah Carroll, Andi Peng, Phillip Christoffersen, **Mehul Damani**, Stewart Slocum, Usman Anwar, Anand Siththaranjan, Max Nadeau, Eric J Michaud, Jacob Pfau, Dmitrii Krasheninnikov, Xin Chen, Lauro Langosco, Peter Hase, Erdem Biyik, Anca Dragan, David Krueger, Dorsa Sadigh, Dylan Hadfield-Menell

Transactions on Machine Learning Research (TMLR)

[6] **Mitigating Generative Agent Social Dilemmas** 2023

Julian Yocum, Phillip Christoffersen, **Mehul Damani**, Justin Svegliato, Dylan Hadfield-Menell, Stuart Russell

NeurIPS 2023 Foundation Models for Decision Making Workshop

[5] **Formal Contracts Mitigate Social Dilemmas in Multi-Agent RL** 2023

Andreas Haupt, Phillip Christoffersen, **Mehul Damani**, Dylan Hadfield-Menell

Autonomous Agents and Multi-Agent Systems

[4] **SocialLight: Distributed Cooperation Learning towards Network-Wide Traffic Signal Control** 2023

Harsh Goel, Yifeng Zhang, **Mehul Damani**, Guillaume Sartoretti

International Conference on Autonomous Agents and Multiagent Systems

[3] **Distributed Reinforcement Learning for Robot Teams: a Review** 2022

Yutong Wang, **Mehul Damani**, Pamela Wang, Yuhong Cao, Guillaume Sartoretti

Current Robotics Reports, Springer

[2] **PRIMAL2: Pathfinding Via Reinforcement and Imitation Multi-Agent Learning - Lifelong** 2021

Mehul Damani, Zhiyao Luo, Emerson Wenzel, Guillaume Sartoretti

IEEE Robotics and Automation Letters

[1] **Flatland Competition 2020: MAPF and MARL for Efficient Train Coordination on a Grid World** 2020

Florian Laurent, Manuel Schneider, Christian Scheller, Jeremy Watson, Jiaoyang Li, Zhe Chen, Yi Zheng, Shao-Hung Chan, Konstantin Makhnev, Oleg Svidchenko, Vladimir Egorov, Dmitry Ivanov, Aleksei Shpilman, Evgenija Spirovska, Oliver Tanevski, Aleksandar Nikov, Ramon Grunder, David Galevski, Jakov Mitrovski, Guillaume Sartoretti, Zhiyao Luo, **Mehul Damani**, Nilabha Bhattacharya, Shivam Agarwal, Adrian Egli, Erik Nygren, Sharada Mohanty

NeurIPS 2020 Competition and Demonstration Track

Projects

Adversarial Vulnerabilities of CLIP

[Repo](#)

- Generated interpretable, universal perturbations that caused misclassification of images by OpenAI's CLIP model.

MA-Sort

[Repo](#)

- Developed multi-agent RL benchmark to test emergence of dominance hierarchies in a group of RL agents.

Vigilant Bot

[Project Page](#)

- Developed RNN-based embedded device for detecting hand gestures and sending automated distress signals.

Vertical Take-off & Landing Aircraft (VTOL)

[Blog](#)

- Designed, assembled and tested an electric Vertical Take-off and Landing aircraft (VTOL) prototype.

Optimal Debris Deorbiting System

[Report](#)

- Conceptualized a method to deorbit space debris from low-earth orbit (LEO) using bidirectional ion thrusters.

Skills

Languages: Python, C, MATLAB

Maching Learning: TensorFlow, Torch, HuggingFace, Ray, WandB

Other: Conda, Docker, Git, Linux, Slurm

Honors and Awards

- Vicom Book Prize, *Top Scorer in Thermodynamics out of 500 students* 2019
- Dean's List, *Top 5% of cohort* 2019, 2020
- Kishore Vaigyanik Protsahan Yojana Fellowship, *Awarded by Department of Science and Technology, India* 2017
- National Talent Search Scholar, *Awarded by National Council on Educational Research, India* 2017
- Invited Tedx speaker on Black Holes and Time Travel 2017

Reviewing

- Transactions on Machine Learning Research (TMLR) 2025
- Conference on Neural Information Processing Systems (NeurIPS) 2022, 2024
- International Conference on Learning Representations (ICLR) 2023, 2025, 2026
- International Conference on Robotics and Automation (ICRA) 2021-2023
- International Conference on Intelligent Robots and Systems (IROS) 2023
- International Conference on Machine Learning (ICML) 2023, 2025, 2026
- Association for the Advancement of Artificial Intelligence (AAAI) 2022
- Robotics and Automation Letters (RA-L) 2022-2024

Invited Talks

- NVIDIA October 2025
- UT Austin October 2025
- CoLM SCALR Workshop Oral October 2025
- MIT AI Conference October 2024

Teaching

- Teaching Assistant for 6.8300 (Advances in Computer Vision) 2024
- Peer Tutor for MA2007 (Thermodynamics) 2021
- Organized tutorial on Arduino-powered robotic cars for high-school students. 2020